B3.3 Muscle and motility

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b3-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 326–340 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Movement

B3.3.1 Motile and sessile organisms

pp. 326–327

All organisms move. **Motile** organisms, such as whales, move from place to place. **Sessile** organisms, such as barnacles and corals, stay fixed but move parts of their bodies to feed, and often have motile larvae.

B3.3.9 Reasons for locomotion

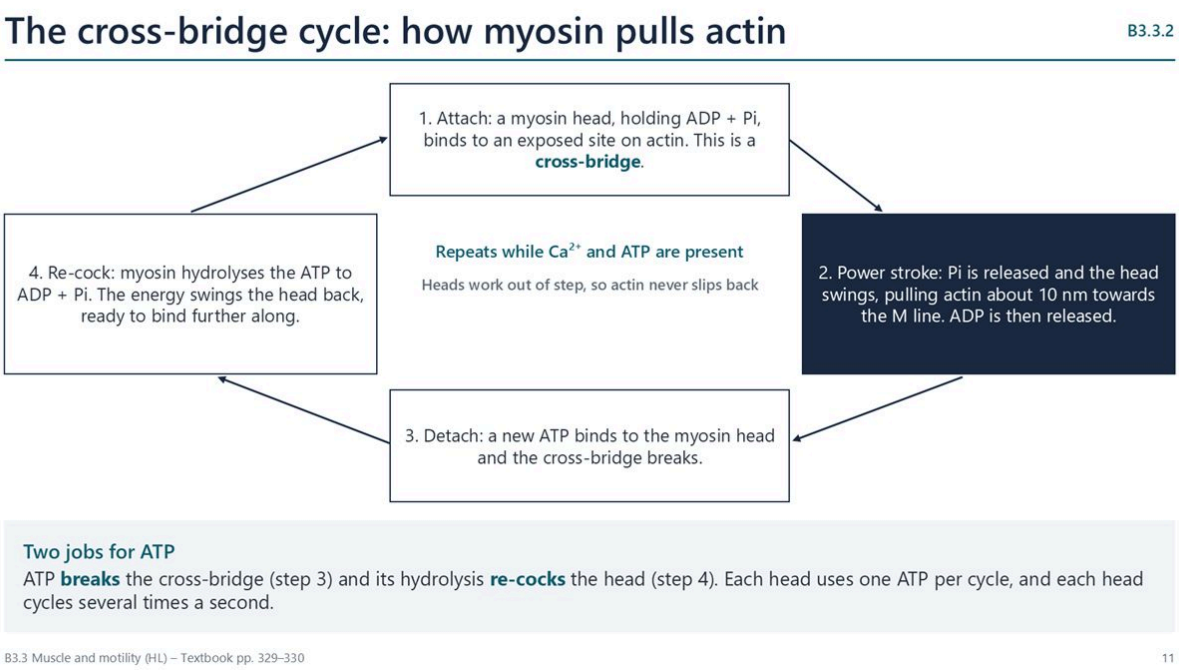
pp. 338–339

Animals move to **forage** (bees collecting nectar), **escape danger** (impala fleeing lions), **find a mate** (male moths following scent) and **migrate** (humpback whales moving between feeding and breeding waters).

Muscle

B3.3.2 Sliding filament model

pp. 327–330



Myofibrils are chains of **sarcomeres** with thick **myosin** and thin **actin** filaments. Calcium ions bind troponin, moving tropomyosin to expose binding sites on actin. Myosin heads bind, swivel and pull actin towards the centre; ATP breaks each cross-bridge and re-cocks the head. Sarcomeres shorten while the filaments stay the same length.

B3.3.3 Titin and antagonistic muscles

pp. 330–331

Muscles only exert force when they contract, so they work in **antagonistic pairs**, each stretching the other. The huge protein **titin** stores energy when stretched, helps the sarcomere recoil and prevents overstretching.

B3.3.4 Motor units

pp. 331–332

A **motor unit** is one motor neuron and all the muscle fibres it controls, connected at neuromuscular junctions. All fibres in a unit contract together. Small units give fine control; using more units gives more force.

B3.3.8 Intercostal muscles

pp. 336–337

The external and internal intercostal muscles have fibres running in different directions, so they move the ribcage in opposite directions. When one layer contracts it stretches the other, storing energy in its titin.

Skeletons, joints and swimming

B3.3.5 Skeletons and levers

pp. 332–334

Arthropods have an **exoskeleton**, with muscles attached inside; vertebrates have an **endoskeleton**, with muscles attached to bones by tendons. Bones act as levers, with the joint as the pivot.

B3.3.6 The hip joint

pp. 334–335

At the hip, the head of the **femur** fits a socket in the **pelvis**. **Cartilage** reduces friction, **synovial fluid** lubricates, **ligaments** join bone to bone, and **muscles** pull on the bones through **tendons**.

B3.3.7 Range of motion

pp. 335–336

Ball-and-socket joints (hip, shoulder) move in three planes; hinge joints (knee, elbow) move mainly in one. Range of motion can be measured with a goniometer and is affected by training, age and injury.

B3.3.10 Marine mammals

pp. 339–340

Whales and dolphins have a **streamlined** body, forelimbs modified as **flippers**, a tail **fluke** moved up and down, and a **blowhole** with an airway separate from the gut, so they can breathe at the surface between dives.

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